

Multi-Priority Cartesian Impedance Control based on Quadratic Programming Optimization

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Abstract—In this work we introduced a prioritized Cartesian impedance control under the framework of the Quadratic Programming (QP) optimization. In particular, we present a formulation which is simpler than full inverse dynamics, avoids any matrix pseudo-inversion, inverse kinematics computation and considers strict priorities among tasks. Our formulation is based on QP optimization permitting to take into account also explicit inequality constraints.

We compare in simulation the tracking results obtained with a classical algebraic implementation against those derived from the proposed QP implementation taking into account joint torque limits. We consider the classical Cartesian impedance controller and a simplified version, also known as *Virtual Model Control*. Finally the proposed method was implemented and validated on a humanoid upper-body torque controlled robot. Experimental trials involving various physical interaction conditions were executed to demonstrate the performance of the proposed method.

I. INTRODUCTION

Robots that have to perform tasks in unknown, real scenarios, need to handle contacts with the environment and the objects to manipulate or interactions with human co-workers. These requirements can be partially addressed by proper hardware (for example Series Elastic Actuators (SEAs) [1]), but a predominant role is constituted by the capability of the controller to handle all these situations. Furthermore, it is important to notice that, in general, the forces involved during all these situations are not known a priori.

Among all possible controllers suitable to tackle the needs of physical interaction, a notable one is the *Cartesian Impedance Controller*, which relates the wrench exchanged by the robot with the environment to Cartesian pose and velocity errors through Cartesian stiffness and damping matrices:

$$\mathbf{f} = \mathbf{K}\Delta\mathbf{x} + \mathbf{D}\Delta\dot{\mathbf{x}}, \quad (1)$$

where $\Delta\mathbf{x}$ is the pose error computed w.r.t. a reference pose and $\Delta\dot{\mathbf{x}}$ is the velocity error w.r.t. a reference velocity. The force $\mathbf{f}$ is then projected into the robot joint torques $\boldsymbol{\tau}$

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through the task Jacobian. Equation (1) permits to handle any kind of interaction by setting different set points and stiffness/damping matrices. Cartesian impedance control has been used for the dynamic interaction between the robot and the environment as well as for motion control.

The contribution of this work is the introduction, implementation and validation of a multi-priority impedance controller that considers task priorities, inside a Quadratic Programming (QP) optimization framework. The main advantage of the proposed formulation of Cartesian Impedance controller is that such an approach allows to cope with highly redundant robots, taking into account hard priorities between tasks, and easily handling constraints in form of equalities and inequalities. In particular, inequality constraints and bounds permit to consider hardware limitations such as joint torque limits that are fundamental when working with real robotic platforms. We show how the proposed formulation is simple and efficient, and we validate the proposed method both in simulations and experiments under physical interactions during motions.

The rest of the article is structured as follows: Section II review the state of art concerning QP-based controllers, Section III describes the details of both the Impedance Control and the Virtual Model Control, Section IV introduces the concept and theory behind our QP-based Impedance Controller, Section V presents the application and validation of our controller in simulated scenarios, Section VI reports experiments on Gazebo and real hardware, and finally Section VII summarizes the main results of this work.

II. RELATED WORKS

Cartesian Impedance Control was firstly introduced by Hogan in [2] as an efficient operational space controller, capable to robustly handle contacts with the environment. The Cartesian Impedance Control does not require the inversion of the kinematics but only the computation of forward kinematics and possibly of the dynamic terms and it requires joint level torque control.

A simplified version of Cartesian Impedance Control, named *Virtual Model Control*, has been presented and implemented on a walking robot by Pratt et al. in [3]. Virtual Model Control consists in a simplified version of Cartesian Impedance Control since it does not require the computation of dynamic terms, but it relies only on the virtual spring-damper control components.

In [4] by Ott et al., Cartesian Impedance Control is extended to the case of redundant robots. In particular, the *Null Space Stiffness* is introduced as secondary task to exploit joint level stiffness in a Cartesian Impedance controller with two priorities. The presented controller can only handle equality constraints.

In [5], Henze et al., applies a hierarchical controller to the humanoid robot TORO for a balancing task using contacts. Cartesian Impedance Control is used to control the end-effectors of the robot. In this work, a QP problem is set up only to find the optimal contact forces distribution but the null space projectors are explicitly computed *outside* the QP itself.

The work in [6], introduces the theory behind a two level Cartesian Impedance controller in which the secondary task does not influence the primary task and it is minimized in accordance with the weighted squared magnitude of the Cartesian acceleration error. Furthermore, they state that the well-known Sentis-Khatib control law of [7] does not guarantee that the secondary task is minimized under the same optimization criterion.

In [8] the inverse dynamics problem is solved in its full extent by formulating a hierarchical QP problem in which torques, accelerations and contact forces are computed all together. On the one hand, such an approach is complete since it encompasses the definition of a tasks in terms of both force and Cartesian position, as well as joint acceleration, torque and contact force constraints. On the other hand, as previously stated, the optimization process involves joint acceleration, joint torque and contact force variables, potentially leading to heavy computation.

The proposed formulation is similar to the reduced form presented in [8], but the method introduced in this work further extends by allowing to handle contacts and motion together using the aforementioned Cartesian Impedance control. Furthermore, the presented method uses a general formulation that is compatible with any QP solver.

III. PROBLEM DEFINITION

Let us consider a fixed-base n -dof manipulator, and let $\mathbf{q} \in \mathbb{R}^n$ be the corresponding joint configuration vector. Moreover, let $\mathbf{x} \in \mathbb{R}^m$ denote a *task*, i.e. some quantity of interest which can be expressed as a non-linear function of the robot configuration

$$\mathbf{x} = \mathbf{x}(\mathbf{q}). \quad (2)$$

Differentiation of (2) yields

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}, \quad (3)$$

where $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{m \times n}$ is the task Jacobian matrix. Since we are mainly interested in the specific case in which the manipulator is redundant with respect to the given task, the inequality $m < n$ will hold from now on.

Through manipulation of the system dynamics equation in contact with the environment (where $\mathbf{f}_{ext}$ are the forces due to the environment):

$$\mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau} + \mathbf{J}^T \mathbf{f}_{ext}, \quad (4)$$

and the task description at the acceleration level (which can be obtained by further differentiation of (3))

$$\ddot{\mathbf{x}} = \mathbf{J}\ddot{\mathbf{q}} + \dot{\mathbf{J}}\dot{\mathbf{q}}, \quad (5)$$

a task-space dynamics equation can be obtained, that is the following:

$$\Lambda(\mathbf{q})\ddot{\mathbf{x}} + \boldsymbol{\eta}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{f} + \mathbf{f}_{ext}. \quad (6)$$

In (6) $\mathbf{f} \in \mathbb{R}^m$ is the vector of task-space forces that are equivalent to the joint torques $\boldsymbol{\tau}$. The relation between these two quantities is linear [9]:

$$\bar{\mathbf{J}}^T \boldsymbol{\tau} = \mathbf{f}, \quad (7)$$

where:

$$\bar{\mathbf{J}} = \mathbf{B}^{-1} \mathbf{J}^T (\mathbf{J} \mathbf{B}^{-1} \mathbf{J}^T)^{-1}, \quad (8)$$

also known as the *dynamically consistent pseudo-inverse* [10]. Given the measured joint torques it is possible, through (7), to compute the force exerted by the robot joints at the end-effector. The inverse problem of (7) is to find the joint torques vector that realizes a certain task-space force. One specific solution of such a problem which is well known is the following.

$$\boldsymbol{\tau} = \mathbf{J}^T \mathbf{f}; \quad (9)$$

The solution from (9) is the one which makes the whole manipulator move as if it was subjected to the force $\mathbf{f}$; however, if we are not interested to this specific property, we have to consider the inverse problem of (7) in its full extent. The general, dynamically-consistent, solution of the inverse problem of (7) can be shown to be:

$$\boldsymbol{\tau} = \mathbf{J}^T \mathbf{f} + (\mathbf{I} - \mathbf{J}^T \bar{\mathbf{J}}^T) \boldsymbol{\tau}_0, \quad (10)$$

in which $\boldsymbol{\tau}_0$ is a vector of torques that generates null-space manipulator motions but no forces (and, consequently, no acceleration) at the end effector.

Let us now move to a slightly more complex case, in which two tasks $\mathbf{x}_1 \in \mathbb{R}^{m_1}$ and $\mathbf{x}_2 \in \mathbb{R}^{m_2}$ are specified, along with their corresponding desired forces $\mathbf{f}_1$ and $\mathbf{f}_2$. Furthermore, let us assume that task 1 is given a higher priority, meaning that we want the robot to "do its best" to perform task 2 *without affecting* the performance achieved at executing task 1. To reach this goal we may exploit the null-space of task 1 as in (10), choosing a vector $\boldsymbol{\tau}_0$ which makes the difference between the desired force $\mathbf{f}_2$ and the actual force $\mathbf{f}_2 = \bar{\mathbf{J}}_2^T \boldsymbol{\tau}$ as small as possible. This can be done via linear algebra techniques [11].

In this work we want to find a way to specify a Quadratic Programming (QP) problem, that solves the inverse problem of (7), taking into account priorities *and* constraints. QP is a convenient way to solve these kind of problems since it permits to specify inequality constraints that are useful in practical implementations. Furthermore, QP has already been used in the domains of Inverse Kinematics (IK) and Inverse Dynamics (ID). In particular, our approach aims at being simpler, but more robust than full inverse dynamics. We consider the desired virtual forces generated by considering virtual components placed at certain locations within the

robot or between the robot and the environment [3], as for example a virtual spring:

$$\mathbf{f} = \mathbf{K}(\mathbf{x}_d - \mathbf{x}(\mathbf{q})), \quad (11)$$

where $\mathbf{K} \in \mathbb{R}^{m \times m}$ is a virtual stiffness matrix, $\mathbf{x}_d \in \mathbb{R}^m$ is the desired pose of the end-effector and $\mathbf{x}(\mathbf{q}) \in \mathbb{R}^m$ is the actual pose of the end-effector.

IV. QP FORMULATION

In this section we are going to explicitly formulate the inverse problem of (7) as a QP problem. More specifically, we want to find a formulation in which the cost functions, as well as the constraints that allow for strict priorities between tasks, are well defined.

A. Single Task

First, notice that we can obtain the control law (9) by considering the following QP problem:

$$\min_{\boldsymbol{\tau}} \|\bar{\mathbf{J}}^T \boldsymbol{\tau} - \mathbf{f}\|^2, \quad (12)$$

in fact, the quadratic cost function of (12) is:

$$F(\boldsymbol{\tau}) = \frac{1}{2} \boldsymbol{\tau}^T \bar{\mathbf{J}} \bar{\mathbf{J}}^T \boldsymbol{\tau} - \mathbf{f}^T \bar{\mathbf{J}}^T \boldsymbol{\tau} + \frac{1}{2} \mathbf{f}^T \mathbf{f}. \quad (13)$$

By convexity of the objective function, the solution has to satisfy the necessary and sufficient unconstrained optimality condition:

$$\frac{\partial F(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}} = \bar{\mathbf{J}} \bar{\mathbf{J}}^T \boldsymbol{\tau} - \bar{\mathbf{J}} \mathbf{f} = 0. \quad (14)$$

If we use (8) in (14), the latter can be written as:

$$\mathbf{J} \mathbf{B}^{-1} \boldsymbol{\tau} = \mathbf{J} \mathbf{B}^{-1} \mathbf{J}^T \mathbf{f}, \quad (15)$$

a solution that satisfies (15) is given by (9), while a parametrization of the general solution is given by (10). The main drawback of formulation (12) is that we have to compute the pseudo-inverse $\bar{\mathbf{J}}$ in order to define the cost function. This is a computationally costly operation that indeed is already handled by the QP solver.

To avoid the computation of $\bar{\mathbf{J}}$, we can use equation (15):

$$\min_{\boldsymbol{\tau}} \|\mathbf{J} \mathbf{B}^{-1} \boldsymbol{\tau} - \mathbf{J} \mathbf{B}^{-1} \mathbf{J}^T \mathbf{f}\|^2, \quad (16)$$

and it can be checked that the family of solutions satisfying (16) is (10), and also (9) as a special case.

B. Prioritized QP Formulation

When multiple tasks are considered together, we can imagine a cascade of QP problems in which, at each level, a new solution $\boldsymbol{\tau}_i$ is found subject to the optimality constraint:

$$\mathbf{J}_{i-1} \mathbf{B}^{-1} \boldsymbol{\tau}_i = \mathbf{J}_{i-1} \mathbf{B}^{-1} \boldsymbol{\tau}_{i-1}^*, \quad (17)$$

in which the optimal torques $\boldsymbol{\tau}_{i-1}^*$ are computed by the previous QP problems.

If we consider the i -th level of priority, the QP problem has the form:

$$\begin{aligned} \argmin_{\boldsymbol{\tau}} \quad & \|\mathbf{J}_i \mathbf{B}^{-1} \boldsymbol{\tau}_i - \mathbf{J}_i \mathbf{B}^{-1} \mathbf{J}_i^T \mathbf{f}_i\|^2 + \varepsilon \|\boldsymbol{\tau}_i\|^2 \\ \text{s.t.} \quad & \mathbf{b}_l \leq \mathbf{A} \boldsymbol{\tau}_i \leq \mathbf{b}_u \\ & \mathbf{u}_l \leq \boldsymbol{\tau}_i \leq \mathbf{u}_u \\ & \mathbf{J}_{i-1} \mathbf{B}^{-1} \boldsymbol{\tau}_{i-1} = \mathbf{J}_{i-1} \mathbf{B}^{-1} \boldsymbol{\tau}_i \\ & \vdots \\ & \mathbf{J}_0 \mathbf{B}^{-1} \boldsymbol{\tau}_0 = \mathbf{J}_0 \mathbf{B}^{-1} \boldsymbol{\tau}_i, \end{aligned} \quad (18)$$

where we are considering a Cartesian task. The second term in the cost function is used as regularisation term, subject to a set of bounds $[\mathbf{u}_l, \mathbf{u}_u]$, constraints $[\mathbf{b}_l, \mathbf{b}_u]$ and optimality conditions given by the previous $i-1$ QPs.

Notice that it is possible to find in literature ways to formalize multiple QP problems in just one QP, for example in [12] and [13].

C. Joint Torque Limits

The joint torque limits are an example of bounds for the QP problem previously formulated. These can be easily written in the form:

$$\boldsymbol{\tau}_{min} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{max}. \quad (19)$$

It is important to notice that any feed-forward term, added to the output torque of the QP, should be subtracted to the available torques. For example, if we consider to compensate gravity and Coriolis-centrifugal terms, the desired joint torques will be:

$$\boldsymbol{\tau}_d = \boldsymbol{\tau}_{opt} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}), \quad (20)$$

where $\boldsymbol{\tau}_{opt}$ are the optimal joint torques computed by the QP Optimization and $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$ are the joint torques to compensate Coriolis-centrifugal terms and gravity respectively. The bounds in (19) should be changed in:

$$\boldsymbol{\tau}_{min} - (\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})) \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{max} - (\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})). \quad (21)$$

The presented formulation permits to easily specify constraints to handle joint velocity and acceleration limits, as in [14]. More examples of possible bounds/constraints can be found also in [11], [8].

V. SIMULATIONS

A. Matlab Simulations

This section introduces a comparison study between our approach, which is based on the formulation of a hierarchical QP, and the classical approach based on the algebraic solution of (8).

We are considering a 6-DOFs, *PUMA560*, arm performing three Cartesian position tasks that consists in:

- 1) reaching 1.0 [m] along the x axis
- 2) reaching 1.0 [m] along the y axis
- 3) following a periodic (0.5 Hz) sinusoidal trajectory, of amplitude 0.3 [m], around the set point 0.0 [m] along the z axis

Part of the resulting reference trajectory is outside the workspace, in particular relaxing the tasks in x and y is possible to fulfill the task in z .

1) *Classical Approach*: First we consider the response of the system considering the simple control law (9), where the forces $\mathbf{f}$ are generated using a virtual spring-damper:

$$\mathbf{f} = \mathbf{K}(\mathbf{x}_d - \mathbf{x}) - \mathbf{D}\dot{\mathbf{x}}, \quad (22)$$

Where $\mathbf{K} = 15000 \left[\frac{N}{m} \right]$ and $\mathbf{D} = 400 \left[\frac{Ns}{m} \right]$ in all the directions.

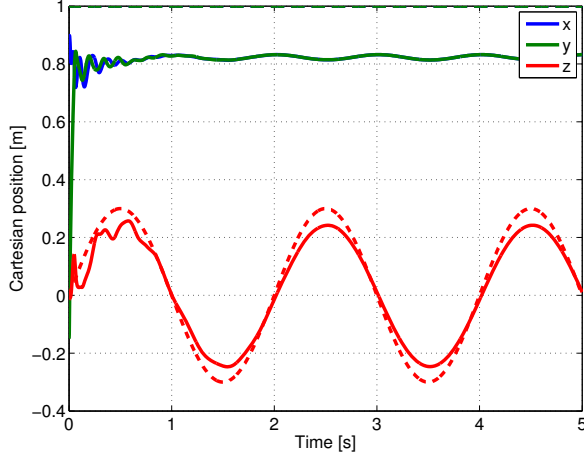


Fig. 1. Cartesian reference trajectories (dashed lines) versus Cartesian computed trajectories (continuous lines) for the control law in (9)

Figure 1 shows that when using the control law in (9) it is not possible to achieve optimality in none of the given tasks. Indeed the tasks in $[x \ y]$ are conflicting with the one in z . Furthermore, the generated joint torques are far too high (due to the chosen high values for stiffness and damping), risking to cause damage to the manipulator. If we want to achieve optimality in one of the given tasks, we need to consider *hard* priorities between them, i.e. by setting the priority of the task's z -component to be higher than the $[x \ y]$ components. To this aim, an algebraic solution is provided by the control law in [11]. In this case two virtual forces will have to be defined: $\mathbf{f}_1$ from the last row of (22) and $\mathbf{f}_2$ from the first two rows of (22). Regarding the high torques, a *naive* way to solve this issue is to implement a simple saturation on the output torques. However, this naive approach does not assure minimum quadratic norm error for the tasks, resulting in large deviations from the desired trajectory (as shown in Figure 2).

Figure 3 shows the Cartesian trajectories obtained when using the control law (10) without joint torque saturation. The large jittering is caused by working near singularities without any damped pseudo-inverse. Finally, it is worth to notice that in all the experiments we were not compensating any dynamics effect.

2) *QP Approach*: We are now going to see the results obtained using the hierarchical QP formulation of (18), using the *qpOASES* [15] library as QP solver. Figure 4 shows that the proposed approach is able to follow the desired high-priority trajectory, minimizing the error norm of the secondary task. Moreover, numerical stability issues related with the reference trajectory being located at the boundary of the robot workspace are handled in a clean way, especially

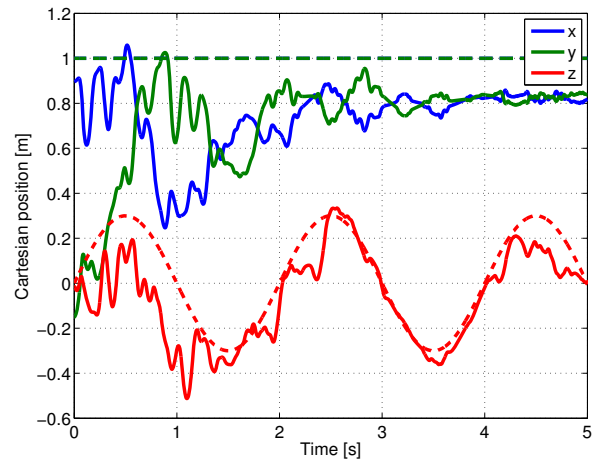


Fig. 2. Cartesian reference trajectories (dashed lines) versus Cartesian computed trajectories (continuous lines) for the control law in (10) including saturation of output torques

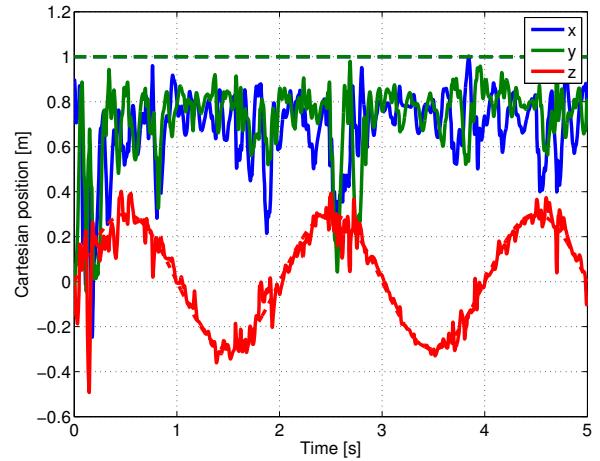


Fig. 3. Cartesian reference trajectories (dashed lines) versus Cartesian computed trajectories (continuous lines) for the control law in (10)

when compared to the outcome of the algebraic approach (see Figure 3). As a final remark, the torque output of this algorithm is guaranteed to always lie inside the constraints, thus allowing to avoid damaging the robot.

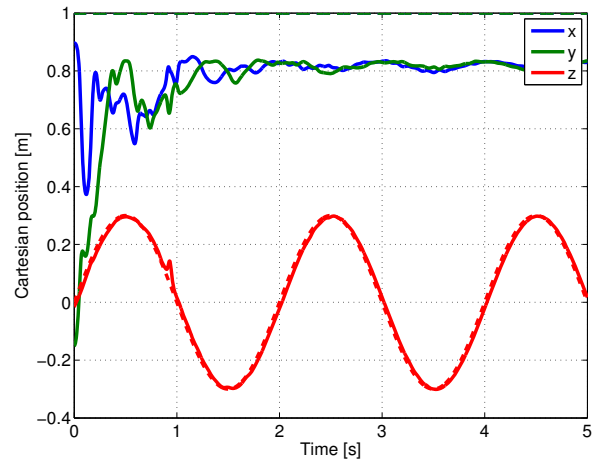


Fig. 4. Cartesian reference trajectories (dashed lines) versus Cartesian computed trajectories (continuous lines) for the control law in (18)

Finally we consider a Cartesian Impedance controller formulated as in [16], the desired joint torques are computed as in (20) and the Cartesian forces are computed as:

$$\mathbf{f} = \mathbf{\Lambda}(\mathbf{q}) (\ddot{\mathbf{x}}_d - \mathbf{J}\dot{\mathbf{q}}) + \mathbf{K}(\mathbf{x}_d - \mathbf{x}) + \mathbf{D}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}), \quad (23)$$

where:

$$\mathbf{\Lambda}(\mathbf{q}) = (\mathbf{J}\mathbf{B}^{-1}\mathbf{J}^T)^{-1}, \quad (24)$$

is the equivalent Cartesian mass matrix, $\mathbf{\Lambda}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$. We are also considering a third level Null Space Stiffness task in the form:

$$\boldsymbol{\tau}_p = \mathbf{K}_p(\mathbf{q}_d - \mathbf{q}) - \mathbf{D}_p\dot{\mathbf{q}}. \quad (25)$$

It is important to notice that the equivalent Cartesian mass matrix (24) can take arbitrarily high values near a singular configuration. For this reason it is normally computed as a *regularized* Cartesian mass matrix. We propose to compute such matrix formalizing a suitable QP problem which is then solved in a numerically-stable way by relying on the *iterative regularization procedure* [17], [15] provided by qpOASES. For the joint torque limits we are using the formulation version in (21).

We are using the same values as before for the Cartesian stiffness and damping, while for the virtual joint stiffness and damping in (25), we are setting $\mathbf{K}_p = 10000 \left[\frac{Nm}{rad} \right]$ and $\mathbf{D} = 100 \left[\frac{Nms}{rad} \right]$ respectively for all the joints. Figure 5 shows

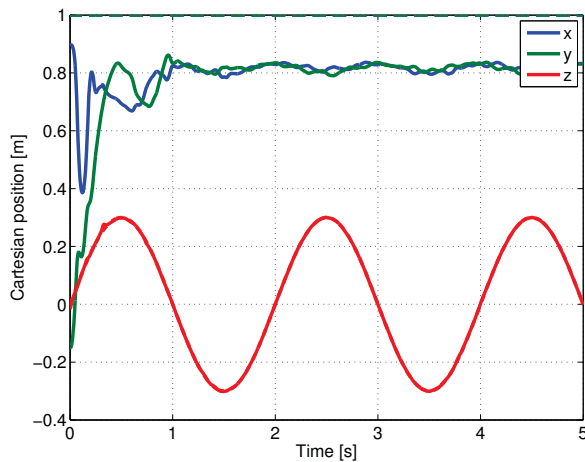


Fig. 5. Cartesian reference trajectories (dashed lines) versus Cartesian computed trajectories (continuous lines) for the control law in (18) considering also (23) and (25)

that the quality of the tracking is better as expected since we are compensating dynamics effects, resulting on the other hand, in a more complicated and less efficient controller. The QP formalization ensures, in both the experiments, that joint torque limits are respected.

VI. GAZEBO SIMULATIONS & EXPERIMENTS

In this section, a series of experiments, using the presented controller formulation, are shown both in Gazebo simulations and real hardware trials. As the previous simulations were showing the capabilities of the presented controller in terms of motion control, here we are also considering interaction

with unknown external forces. We choose a humanoid upper-body robot that consists in a two 7-DOFs arms mounted on top of a 1-DOF torso, as shown in Figure 6 [18]. The platform relies on torque control for the torso joint and the first four joints of the arms chains. The wrist joints are controlled in position mode and set to a fixed position value. For this reason we consider the Cartesian position part of the end-effectors pose and we do not use the wrist joints in our controller. Therefore we are considering two kinematics chains, starting from the torso to the end-effectors, both sharing the torso DOF.

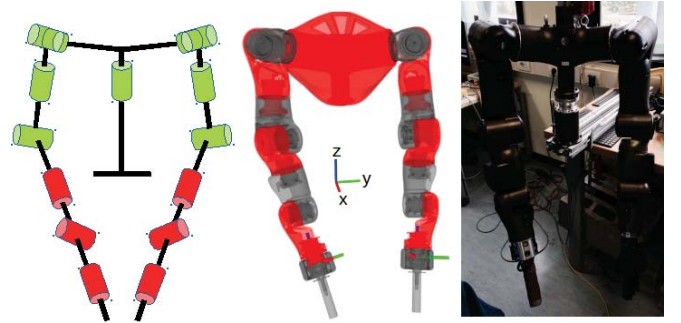


Fig. 6. The green joints are controlled in torque mode while, the red ones are controlled in position mode and set to a fixed position values (left picture); the middle picture shows the frames used for the control; the right picture shows the real hardware platform

Our controller consider two level of priorities:

- 0) Left and right arm Cartesian control (together through task augmentation)
- 1) Null-Space Stiffness task (25)

where 0 is the highest priority, subject to joint torque limits. The forces to generate the motion of the arms in Cartesian space is obtained using the simplified law in (22). We are not completely compensating all the acceleration related dynamical effects of the motion. Coriolis, centrifugal and gravity terms are compensated as in (20), therefore, the joint torque limits are set using (21). For the experiments presented in this paper, the Cartesian impedance for the left and right arm is set to $\mathbf{K} = 700 \left[\frac{N}{m} \right]$ and $\mathbf{D} = 70 \left[\frac{Ns}{m} \right]$ in all the directions. The Null-space stiffness is set to $\mathbf{K}_p = 5 \left[\frac{Nm}{rad} \right]$ and $\mathbf{D}_p = 2 \left[\frac{Nms}{rad} \right]$ to all the joints.

The controller is written in C++, inside the OpenSoT [19], [20] library and the *inequality Hierarchical QP* [21] is implemented using the qpOASES [15] solver. The controller is executed under the XBotCore framework [22], in real time (using Xenomai 2.6), within a control loop running at 1 kHz.

A. Gazebo: Torque Limits

In this experiments, the torque limits of all the joints are restricted to $\pm 20 \left[Nm \right]$. The simulated robot has to keep its end-effectors pose while an external force of $20 \left[N \right]$ is applied, for two seconds, at the right end-effector. Figure 7 shows how the generated torques are bounded inside the given joint limits and thanks to the QP formulation, an optimal solution given the constraints is found. When the

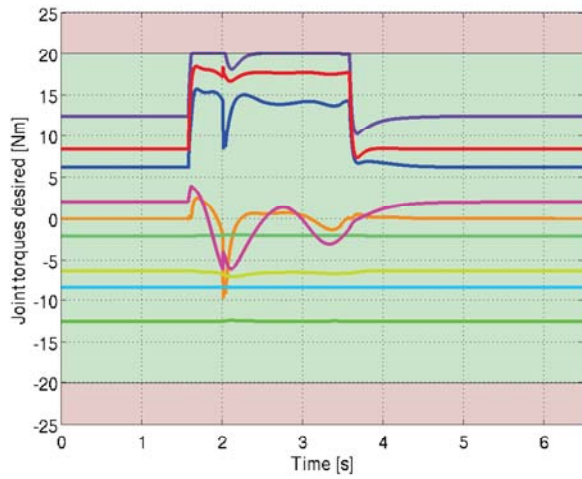


Fig. 7. Joint torques computed by the controller: the green area represent the admissible region for the joint torques, the red area are the bounds

external force is not applied anymore, the end-effector goes back to its previous position.

B. Gazebo: Cartesian Circular Trajectory

In the second Gazebo experiment, the left arm of the robot is commanded to execute a circular trajectory while the right arm has to keep the end-effector pose steady. Multiple external forces are applied to the end-effectors and the structure of the robot, as shown in Figure 8. In particular, external forces are applied for 2 [sec] along the torso (30 [Nm]), the left elbow (50 [N]), the right end-effector (30 [N]) and the left end-effector (30 [N]) consecutively. Figure 9 shows the desired Cartesian trajectories for the left

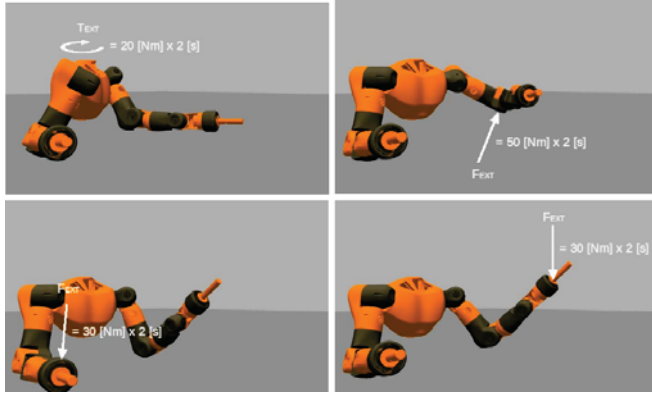


Fig. 8. During the Gazebo simulation, the robot end-effector and structure are disturbed by unknown high external forces

end-effector along the x , y and z axis w.r.t. the computed ones. As expected, when the left end-effector is not directly perturbed, the external forces applied to the structure of the robot does not affect the motion result.

C. Robot: Cartesian Circular Trajectory

The previous experiment is replicated now in the real robotic hardware. Figure 10 shows where the external forces were applied during the motion.

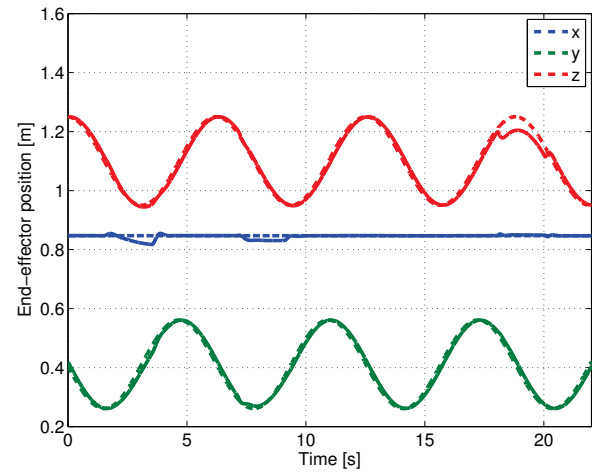


Fig. 9. Desired (dashed) versus computed (continuous) Cartesian trajectories for the left end-effector under external pushes in the Gazebo experiment

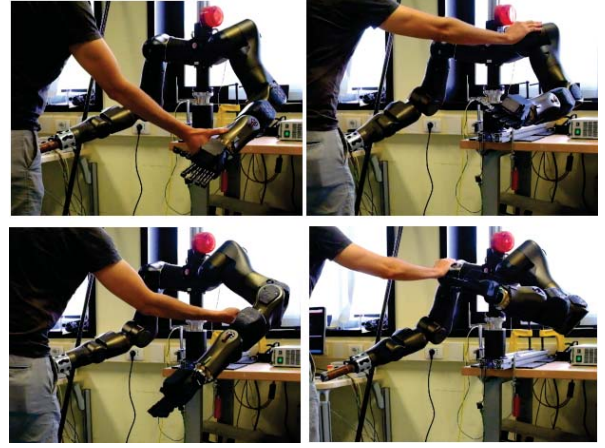


Fig. 10. During the experiment the robot end-effector and structure are disturbed by unknown high external forces

Figure 11 shows the comparison between desired and computed Cartesian trajectories for the left end-effector. As expected, there is a small error due to the uncompensated dynamical effects, the imperfect torque tracking (Figure 12) and the unmodeled dynamics. In this regard, it is worth to point out that we relied on CAD-based inertial properties in order to compute the robot dynamics, without any further identification procedure. Despite this, we can see a fairly good tracking quality and interaction behaviour. As in the previous case, larger error tracking are present when the end-effector is directly disturbed while, when the structure is disturbed, the tracking quality remain the same, thanks to the imposed priorities.

VII. CONCLUSIONS

In this paper we formulated a new prioritized Cartesian Impedance control under the QP optimization framework. The formulation avoids any pseudo-inverse and Inverse Kinematics step and can be used to efficiently cope with motion and interaction. Furthermore this permits to restrict the number of variables to only torques w.r.t. the classical Operational

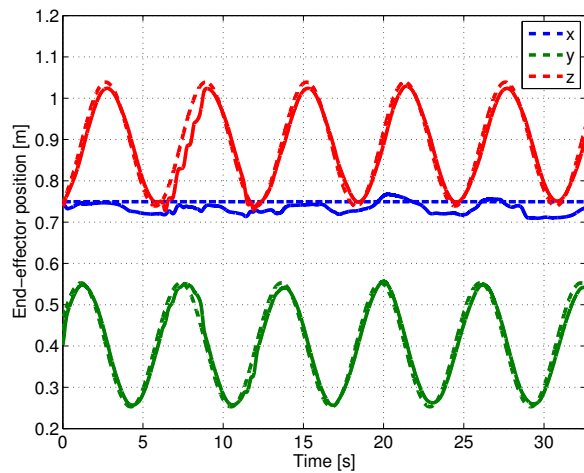


Fig. 11. Desired (dashed) versus computed (continuous) Cartesian trajectories for the left end-effector under external pushes in the real hardware experiment

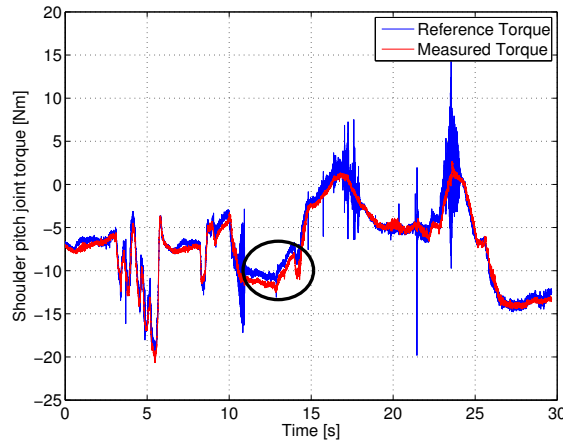


Fig. 12. Comparison between measured torques and references, computed by our algorithm, for the left shoulder pitch joint. It can be notice that there is, in some situation, a tracking error of approximately 10% as shown in the black circle

Space approaches. We considered simple joint torque limits in order to show the application of our formulation in Matlab simulation in which we control a 6-DOFs arm performing position tasks with specified priorities. We also presented results in simulation and on a real robotic platform, constituted by two 7-DOFs arms and a 1-DOF torso, considering tasks with high level of interaction between the human and the robot. Results demonstrated the better performance of the proposed QP formulation against the classical approaches considering also the handling of constraints. Future works will consider the introduction of new tasks and constraints, the floating base formulation and the evaluation of adaptation laws for the Cartesian impedance as the one in [23].

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